

Apostol, Chapter 1 - Exercise 17

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August 2026

17. Prove that if $2^n + 1$ is prime, then n is a power of 2.

Suppose that n is not a power of 2, that is, $n = 2^k m$, where $m > 1$ is odd and $k \geq 0$.

For any odd $j \geq 1$ we have the factorization,

$$x^j + 1 = (x + 1)(x^{j-1} - x^{j-2} + x^{j-3} - \dots - x + 1).$$

Taking $x = 2^{2^k}$ and $j = m$, we get

$$2^n + 1 = 2^{2^k m} + 1 = (2^{2^k} + 1)(2^{2^k(m-1)} - 2^{2^k(m-2)} + 2^{2^k(m-3)} - \dots - 2^{2^k} + 1).$$

Obviously, $2^{2^k} + 1 > 1$, since $k \geq 0$.

The second factor is also bigger than 1. To see this we can agrupate the terms in the following way:

$$(2^{2^k(m-1)} - 2^{2^k(m-2)}) + (2^{2^k(m-3)} - 2^{2^k(m-4)}) + \dots + (2^{2^k \times 2} - 2^{2^k}) + 1$$

every bracketed () expression is bigger than 0, thus the expression above is bigger than 1.

Thus, $2^n + 1$ is composite (contradiction).

Therefore, n is a power of 2.